

A device type definition SHALL NOT alter the choice sets of cluster elements in a way that would be incompatible with the cluster definition.

Examples of choice set changes:

Cluster Choice Set	Change by Device Type Def.	Allowed?	Reason
O.a	O.a+	No	Client expects mutual exclusion
O.a+	O.a	Yes	Subset of client expectations
O.a3+	O.a4+	No	Superset of client expectations
O.a4+	O.a3+	Yes	Subset of client expectations

A device type definition SHALL specify a device type ID, device revision, and a set of one or more mandatory clusters including each cluster's minimum revision.

A device type definition MAY be generic and allow many similar clusters, where at least one instance SHALL be required.

For example: a simple sensor device.

If all sensor devices are common in cluster requirements (except the clusters that perform the sensing), then there is no reason to create a device type for each sensor cluster.

A device type definition MAY be very specific and list particular clusters as mandatory.

For example: a door lock device or thermostat.

7.16.1. Device Type Revision

A device type revision is an unsigned integer that is associated with an approved revision and release of a device type definition. The initial value for a device type revision SHALL be 1. The initial revision (1) of a device type definition SHALL require the latest (at the time of definition the cluster) certifiable revisions of the clusters it mandates. Any mandatory changes to the device type definition SHALL only augment, not modify, the function of the device. Any changes SHALL increment the version of the device. Newer versions of the device SHALL interoperate with older revisions at the older revision's level of functionality.

Examples of changes to a device type definition that require incrementing the revision:

- Mandating a higher revision of one or more mandatory clusters